BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: 102101

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. If the pendulum completes exactly 12 cycles in 2.0 min, what is the frequency of the pendulum?

(A) 0.10 Hz
(B) 0.17 Hz
(C) 6.0 Hz
(D) 10 Hz

Q2. Various instruments sound different even they play the same note due to difference in _____.

(A) Loudness
(B) Pitch
(C) Intensity
(D) Quality

Q3. Which one of the following will refract light more?

(A) Glass
(B) Air
(C) Diamond
(D) Water

Q4. Radius of curvature and focal length for convex mirror are related by equation _____.

(A) f = 2R
(B) f = 1/2 R
(C) f = - 1/2 R
(D) f = -2R

Q5. The value of coulomb constant "K" depends on.

(A) Charges
(B) Separation between charges
(C) Material medium
(D) All of these

Q6. The device that is used to protect a circuit against overload is _____.

(A) Fuse
(B) Lamp
(C) Heater
(D) All of these

Q7. Batteries are rated with unit "ampere-hour" it is the unit of _____.

(A) Current
(B) Power
(C) Charge
(D) Energy

Q8. Which derived unit is equivalent to tesla T.

(A) Nm/A
(B) NA/m
(C) N/Am
(D) Am/N

Q9. The Boolean equation for NAND gate is _____.

(A) X=AB
(B) X=A+B
(C) X=AB with bar over (AB)
(D) X=A+B with bar over (A+B)

Q10. Cathod rays are beam of.

(A) Electrons
(B) Protons
(C) Neutrons
(D) Ions

Q11. If the angle of incidence is 30° the angle of reflection will be.

(A) 30°
(B) 45°
(C) 90°
(D) 210°

Q12. What type of nuclear decay leaves the number of protons and neutrons unchanged?

(A) Alpha decay
(B) Beta decay
(C) Gamma decay
(D) Both A and B

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) Prove the relation between wave speed, wave length and frequency of wave.

(ii) What is the speed of sound in air at -20°c?

(iii) Are rearview mirrors used in cars concave or convex?

(iv) Define capacitance of a capacitor. Write its mathematical form and unit.

(v) A car has two headlight, when the filament in one headlight burns out, the other headlight stays on. Are the head lights connected in series or parallel?

(vi) While starting an engine of a truck, its battery sets 720C of charge in motion for 4.0S. How much current is flowing?

(vii) Can an electron at rest be set into motion with a magnetic field?

(viii) A constant magnetic field is applied to a current carrying conductor. What angle should the wire make with field for the force to be (a) maximum (b) minimum?

(ix) What are the Boolean equation to represent the output of OR, AND, and NOR gates.

(x) What is meant by reflection of light? State laws of reflection of light?

(xi) Lead-210 has half life of 22.3 years. How much of the 80mg of lead will be left after 66.9 years?

SECTION "C"

Maximum Marks: 21

> Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [4] What is audible frequency range. Differentiate between infra-sound and ultra-sound.

(b) [3] Show that the mass spring system execute simple harmonic motion.

QIV.

(a) [4] Derive thin lens Equation.

(b) [3] How much work must be done in moving a charge of 3 coulomb from point at 118V to a point 138V in an electric field.

QV.

(a) [4] Define electric current. In what units it is measured? Discuss briefly the direction of current through the conductor.

(b) [3] A certain transformer has a primary winding with 500 turns and a secondary winding with 250 turns. What is the turns ratio?

QVI.

(a) [4] What are radioisotopes? Explain their uses for various applications?

(b) [3] Define logic gates. Describe the operation of NAND and NOR logic gates by drawing their symbols and truth tables.